

VINIT KUMAR

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EDUCATION

Maharshi Dayanand University — *B.Tech in Computer Science and Engineering*
Rohtak, Haryana

2024 – 2028

FEATURED PROJECTS

g8g — Workflow Automation Platform *(Personal Project)*

2025 – Present

github.com/VinitKumar01/n8n-clone • g8g-web.vercel.app — Go, Next.js, TypeScript, PostgreSQL, ReactFlow, Docker, Tailwind CSS

- Built a full-stack visual workflow automation platform (n8n-inspired) from first principles with a high-performance Go backend and a Next.js + ReactFlow interactive canvas.
- Designed and implemented a custom **DAG execution engine in Go** with topological sorting, Tarjan cycle detection, and a per-node goroutine worker pool with in-degree tracking to safely execute parallel branches.
- Engineered multi-trigger support: manual UI execution, **webhook ingestion** with runtime dynamic route registration, and interval schedulers using Go tickers with UUID job tracking.
- Integrated Google Gemini AI prompting, Resend email dispatch, and fan-in merge nodes into a centralized executor registry; persisted workflow schemas in PostgreSQL using type-safe SQLC queries.

Astrologer Pankaj Bahl Platform *(Client Project)*

2025

astrologer-web.pages.dev — Hono.js, Cloudflare Workers, Vite, React 19, TypeScript, PostgreSQL, Prisma, Upstash Redis, Razorpay

- Engineered an edge-native production consulting, Vedic puja booking, and e-commerce platform for Astrologer Pankaj Bahl deployed on Cloudflare Workers (Hono.js) and Cloudflare Pages.
- Architected **two-tier caching** (in-memory L1 Map + L2 Upstash Redis with a 400ms timeout race fallback and non-blocking writes) for sub-second catalogue and horoscope delivery.
- Enforced single-model atomic write invariants and an in-memory **ConcurrencyLimiter (max 6 active queries)** to eliminate connection permit exhaustion on serverless PostgreSQL.
- Integrated Razorpay Orders and Invoices API with automated Resend transactional email tax receipts (PDF generation) and Twilio WhatsApp notifications across a 17-tab administrative portal.

TruckSpark — Vehicle Showcase Platform *(Client Project)*

2024

github.com/VinitKumar01/truck-spark — Next.js, TypeScript, Express, PostgreSQL, Prisma, Redis, AWS EC2, Docker, Clerk, UploadThing

- Delivered an end-to-end vehicle showcase and side-by-side spec comparison platform for a commercial automotive client with an audience of over **58,000 YouTube subscribers**.
- Engineered vehicle management system with multi-criteria search, side-by-side spec comparison, and role-based administrative dashboard using Clerk authentication.
- Built Express backend with **Redis in-memory caching**, delivering sub-50ms query responses for vehicle comparisons and reducing database load by over 70%.
- Diagnosed AWS EC2 boot initialization race conditions and wrote a custom **Linux systemd service** with automated watchdog recovery, sustaining 100% uptime over 4 months of production.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL

Frameworks & Web: Next.js, React, Node.js, Express, Hono.js, ReactFlow, Tailwind CSS, WebSockets

Databases & Cache: PostgreSQL, Redis, MongoDB, Prisma ORM, SQLC

Cloud & DevOps: Cloudflare Workers & Pages, Docker, AWS EC2, Linux, Systemd, Git, Neovim, REST APIs